EFFECTS OF EL NIÑO

- Delayed onset of the rainy season
- Longer or extended dry season
- Fewer tropical cyclones enter the Philippine Area of Responsibility
- Below normal rainfall, which may lead to drought
- Warmer air temperature
- Shortage in water and food supply

SECTORAL IMPACTS OF EL NIÑO

AGRICULTURE

- Decline in soil health and crop production
- Increase in pests and diseases

MARINE RESOURCES

- Fish kills and red tide
- Decline in fish production

WATER RESOURCES

- Reduced streamflow and groundwater due to less rainfall
- Less water supply and quality for dams, irrigation, and power generation
- Over-extraction of groundwater

HUMAN HEALTH

- Less water for hygiene and sanitation
- Higher cost of drinking water due to reduced supply
- Increase in tropical, pulmonary, and food and water-borne diseases

ENVIRONMENT

- Land or soil degradation
- Increase in forest or peatland fires
- Poor air quality

METEOROLOGICAL DRY SPELL/ DROUGHT CONDITIONS

DRY CONDITION	DRY SPELL	DROUGHT
Below Normal Rainfall Conditions for **2 Consecutive Months**	**Below Normal** Rainfall Conditions for **3 Consecutive Months**	**Below Normal** Rainfall Conditions for **5 Consecutive Months**
	Way Below Normal Rainfall Conditions for **2 Consecutive Months**	**Way Below Normal** Rainfall Conditions for **3 Consecutive Months**

**Near normal rainfall conditions 81%- 120%*
***Below normal rainfall conditions = 21% - 60% reduction from average;*
Way below normal rainfall coverage = More than 60% reduction from average

CONTACT US

(02) 8284-0800 loc. 906
information@pagasa.dost.gov.ph
pagasa.climps@gmail.com
bagong.pagasa.dost.gov.ph
PAGASA Science Garden, BIR Road, Brgy. Central, Quezon City, Metro Manila 1100
@pagasa.dost.gov.ph
@dost_pagasa
DOST-PAGASA Weather Report
@dost_pagasa

Produced by **Project #MAGHANDA**

Funded by DOST-GIA, this project is implemented by DOST-PAGASA with DOST-PHIVOLCS and DOST-STII, in partnership with DILG and DILG-LGA. DOST-PCIEERD monitors the project.

Printed December 2022

Scan the QR code to access DOST-PAGASA's website

#MAGHANDA
Meteorological And Geological Hazard Advisories, Warnings and Notifications for Decisive Action

EL NIÑO

AWARENESS AND PREPAREDNESS

DOST | DOST-PAGASA | DOST-PHIVOLCS | DOST-STII | DOST-PCIEERD | DILG | DILG-LGA

What is ENSO?

The **El Niño Southern Oscillation (ENSO)** is a naturally occuring phenomenon of the climate system resulting from **the interaction between the ocean and atmosphere** in the Central and Eastern Equatorial Pacific (CEEP).

El Niño *is the warm phase of ENSO.*

Lasts 8-12 months. Occurs every 2-7 years.

Source: CPC, NOAA

YOU KNOW IT'S EL NIÑO WHEN THERE ARE...

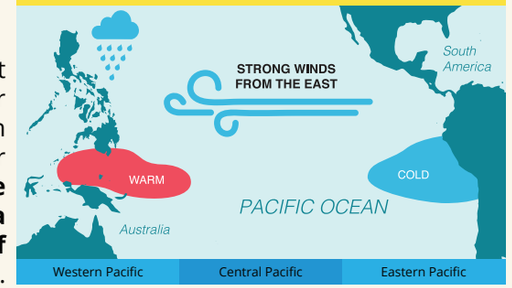
WARMER TEMPERATURES
+0.5 °C or higher than average at the CEEP.

WEAK WINDS FROM THE EAST
that can reverse to become winds from the west.

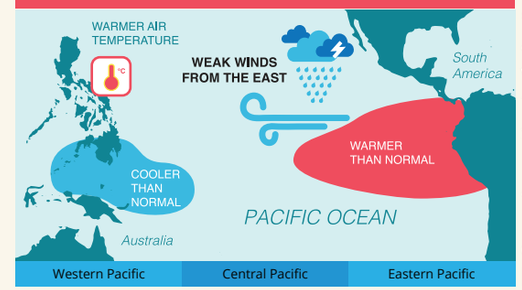
FEWER CLOUDS OVER THE PHILIPPINES therefore, less rain.

Winds from the east push warm water towards the Western Pacific. Clouds form over the Philippines, so **the country receives a normal amount of rainfall.**

PACIFIC OCEAN DURING ENSO-NEUTRAL



PACIFIC OCEAN DURING EL NIÑO



Winds from the east weaken, and are unable to push warm water towards the west. Warm water scatters over the Pacific. Clouds form away from the Philippines, so **the country receives less rainfall.**

ENSO ALERT AND WARNING SYSTEM (EL NIÑO)

The ENSO Alert and Warning System **aims to heighten awareness about El Niño and La Niña events in the country**. It serves as a guide for applying the El Niño/ La Niña Contingency Plan, a program that aims to **prepare Filipinos for the effects of these climate patterns**.

The PAGASA ENSO Alert and Warning System is adopted to provide guidance in the preparation of sectoral contingency plans that will serve as the basis for actions before, during, and after El Niño, as prescribed in Memorandum Order No. 38, (s.2019) of the office of the President.

Reactivating and reconstituting the El Niño Task Force created pursuant to Executive Order No. 16 (s. 2001) and for other purposes.

WARNING TYPE	WATCH	ALERT	ADVISORY	FINAL ADVISORY
	BE AWARE and PREPARED!	*EARLY ACTION!*	*TAKE ACTION!*	*ASSESS AND ACT WHENEVER NECESSARY*
FORECAST	If the conditions are favorable with **55% chance of El Niño development** within the **next six months**	If the conditions are favorable with **70% chance of El Niño development** within the **next two months**	El Niño is **ongoing and expected to continue**.	**El Niño has ended.**
FORM OF ISSUANCE	• Monthly Climate Assessment and Outlook • Press Statement	• Monthly Climate Assessment and Outlook • Press Statement • Dry Spell and Drought Assessment and Outlook	• El Niño Advisories • Press Statement and Press Briefing • Dry Spell and Drought Assessment and Outlook	• Final Advisory • Press Statement • Dry Spell and Drought Assessment and Outlook

SOME TIPS ON WHAT TO DO

STAY UPDATED on climate advisories.
COOPERATE on local El Niño measures.
PREPARE your Go Bag containing items needed for survival

CONSERVE water.
PROVIDE financial assistance in terms of loans.

FIX OR REPORT water leaks.

CREATE a list of seed reserves and planting materials.

MAKE a list of all agricultural machinery and equipment that is operational.

ADVISE farmers to get crop insurance.

PROMOTE integrated pest management.

GET crop insurance.

IMPLEMENT recovery-related projects, programs, & activities.

DISTRIBUTE seed reserves & planting materials.

DISTRIBUTION of agricultural machinery & equipment.

ASSESS damage & losses.

Be aware and prepared — *Early action* — *Take action* — *Final or finish line*

The above actions can be compared to runner's preparations before a race.